A LATAM HACK DAYS WORKSHOP

Open Tools, Open Sky: Doing LSST Science with Public Rubin Data & LINCC Frameworks

Three days turning open access to Rubin's alert stream into working, broker-served science.

📍 Universidad de Antioquia · Medellín, Colombia

🗓 October 13–15, 2026 · 3 days, in person



Goals

- **Turn public access into practical capability.** Enable LATAM researchers to do science with Rubin alert products, even without proprietary data rights.
- **Build hands-on expertise.** Onboard participants to community brokers and LINCC Frameworks: LSDB/HATS, Hyrax, and LightCurveLynx.

- **Create sustained mentorship.** Pair each team with a Rubin mentor and support follow-up toward publications and long-term collaborations.

- **Share reusable resources.** Develop a public toolkit and a “Doing LSST Science without Data Rights” playbook.

Target: ~25 participants • 5 Rubin mentors • 3-day in-person workshop in Medellín

Methodology

BEFORE THE WORKSHOP

- **Mentor kickoff.** Each team agrees on a science question and divides initial tasks.

DURING THE 3-DAY SPRINT

- **Short context talks.** Rubin public-data policy, alert brokers, and Science Pipelines orientation.
- **Team-based hack sprints.** Four ~3-hour sprints, daily stand-ups, rapid updates, and final show-and-tell.

WORKING PRINCIPLES

- **Hands-on first.** Talks take no more than 25% of contact time; most work happens in teams.
- **Mentored and reproducible.** Each team uses at least one community broker and one LINCC Frameworks tool.

Expected results

- **Working science pipelines.** Teams leave with functioning workflows built around public Rubin alerts and LINCC Frameworks.
- **More inclusive participation.** Researchers without LSST data rights gain practical entry points to Rubin science.
- **Pathways to publications.** Mentor follow-up helps teams turn sprint outcomes into manuscripts and ongoing collaborations.
- **Stronger community engagement.** Participants are supported in joining Rubin Science Collaborations that do not require data rights.

Deliverables

- **Team GitHub repositories.** Five mentor-led repositories with Jupyter notebooks and documentation.
- **Workshop repository.** Tutorials, agenda, recordings, and container image, linked from the LSST-DA website.
- **“Doing LSST Science without Data Rights” playbook.** Reusable agenda, application form, container image, broker-selection worksheet, collaboration templates, and lessons learned.
- **Mentor follow-up.** Three monthly check-ins per team, with notes published in team repositories.
- **Evaluation and community outcomes.** Entry/exit surveys and a lessons-learned report; target of ≥8 participants joining an LSST Science Collaboration within 6 months.

WORK TEAMS:

TEAM 1 — MENTOR: ALEJANDRA MUÑOZ ARANCIBIA

Maryory Loaiza Agudelo
Universidad de Antioquia, Colombia

Diego Fernando Fonseca Moreno
U. Nacional de Colombia / U. Antonio Nariño, Colombia

David Sebastián Rodríguez Numpaque
Universidad de Antioquia, Colombia

Sergio Andrés Sotovando Ramírez
Universidad del Valle de Guatemala, Guatemala

Marcos Alejandro Tello Sánchez
Universidad Autónoma de Chiriquí, Panamá

TEAM 2 — MENTORS: BOLIVIA CUEVAS-OTAHOLA & TATIANA ACERO-CUELLAR

Camilo Delgado-Correal
Universidad Distrital Francisco José de Caldas, Colombia

Carlos Mario García Álvarez
Universidad de Antioquia, Colombia

Luis Daniel Díaz Durango
Universidad de Antioquia, Colombia

Franchesca Mabel Del Orbe Acosta
U. Autónoma de Santo Domingo, Dominican Republic

Iván Gabriel Mafla Bolaños
U. Internacional de La Rioja, Spain / Ecuador

TEAM 3 — MENTOR: YIJUNG KANG

Luz Ángela García
Universidad ECCI, Colombia

Juan Camilo Sánchez
Universidad Nacional de Colombia, Colombia

Alexandra Serrano Mendoza
Universidad Industrial de Santander, Colombia

Holman Daniel Quintero Salazar
Universidad de Guanajuato, Mexico

Simon Rodriguez
Universidad de Antioquia, Colombia

Juan Carlos Muñoz-Cuartas
Universidad de Antioquia, Colombia

TEAM 4 — MENTOR: BOLIVIA CUEVAS-OTAHOLA

Verónica Arias Callejas
Universidad Nacional de Colombia, Colombia

Daniel Giovanny Vargas Wilches
Universidad Nacional de Colombia, Colombia

Nil Andy Armillon Huamán
Universidad Nacional de Ingeniería, Peru

Daniel Alfredo Rodríguez De Sousa
Universidad Central de Venezuela, Venezuela

Vanessa Piedrahita Castro
Universidad de Guanajuato, Mexico

TEAM 5 — MENTOR: PEDRO BERNARDINELLI

Jorge Iván Zuluaga Callejas
Universidad de Antioquia, Colombia

Juanita Andrea Agudelo
Universidad de Antioquia, Colombia

Angie Melissa Gutiérrez Estupiñán
Universidad Industrial de Santander, Colombia

Nicolás Pan
Universidad de la República, Uruguay

Valeria Judith Burgoa Camacho
Universidad Mayor de San Andrés, Bolivia

Pedro Bernardinelli ↗ MENTOR

- Exploring Rubin Solar System discoveries in the Minor Planet Center.

Team repository:
bernardinelli/medellin_workshop_solarsystem ↗

Bolivia Cuevas-Otahola ↗ MENTOR

- Mapping the Nearby Universe: distance determination with variable stars in the Rubin alert stream.
- Discovering the Unexpected: anomaly detection in Rubin alert streams.
  Co-mentored with Tatiana Acero-Cuellar ↗

Team repositories:
taceroc/LATAM_Hackathon_AnomalyDetection_2026 ↗
umbramortem/open_tools_open_sky_LSST_2026 ↗

Yijung Kang ↗ MENTOR

- Matching host galaxies of SNe Ia using LSST alerts and archival catalog data.

Team repository:
ykangastro/Team3_SNe_Host_Matching ↗

Alejandra Muñoz Arancibia ↗ MENTOR

- Characterizing transients, AGN and variable stars in Rubin alerts using the ALeRCE broker.

Team repository:
ale-munozarancibia/LSST_Hackdays_Medellin_Team1_Characterizing_DIA Objects ↗

Please remember, this is not a competition, goals are already clear!

Learn to use public alerts, create community, “spread the word” on the public alerts of LSST and available software tools….

Code of Conduct

All participants are expected to follow the LINCC Frameworks Code of Conduct and the LSST-DA Code of Conduct throughout all workshop activities, online and offline.

- **We pledge to treat all people with respect** and provide a harassment- and bullying-free environment, regardless of personal attributes.
- **We pledge to respect the work of others** by recognizing acknowledgment and citation requests of original authors.
- **We pledge to welcome questions and answer them respectfully**, paying particular attention to those new to the community.
- **We pledge to be conscientious of the perceptions of the wider community** and to respond to criticism respectfully.
- **We pledge to help the entire community follow the code of conduct** and not remain silent when we see violations.

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